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PGA-UNet: A Lightweight Prompt-Guided Attention U-Net for Interactive Bone X-Ray Lesion Segmentation

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ABSTRACT Segmenting bone lesions in X-ray images can be difficult when lesions occupy a small image area or have low contrast. We present PGA-UNet, a compact convolutional network for segmentation from an image and a lesion-covering box. A Gaussian-smoothed plateau encodes the box and conditions encoder features through a Prompt Spatial Gate and decoder skip attention through a Conditional Attention Decoder. We evaluate separate models on BTXRD and FracAtlas using image-level splits and simulated boxes under centered covering and predefined off-center conditions. At 512×512, PGA-UNet obtains image-level merged Dice scores of 0.782 and 0.727 under covering prompts, compared with 0.760 and 0.593 for a U-Net baseline with a binary prompt channel. Under the tested off-center condition, its Dice scores are 0.774 and 0.713. These comparisons are descriptive point estimates from the fixed test splits. Ablations show dataset-dependent effects of the conditioning components, while four additional splits characterize variability for PGA-UNet alone. The model contains approximately 2.96 million parameters. The findings support PGA-UNet as a compact design for controlled box-prompted segmentation. Evaluation assumes a correctly localized lesion and does not establish performance with clinician-drawn or off-target prompts.

INDEX TERMS bone X-ray, interactive segmentation, medical image segmentation, prompt-guided learning, Attention U-Net

I. INTRODUCTION

THANKS to its low cost, speed, and accessibility, bone X-ray imaging is widely used in screening, diagnostic support, and treatment monitoring. However, segmenting lesions in this type of image is not simple when the lesion is very small, low-contrast, or partially obscured by the surrounding bone structure. Bounding-box prompts provide an explicit location cue for segmentation. This study examines segmentation conditional on such a cue, using boxes simulated from annotated lesions. The task therefore assumes prior localization of the target region and evaluates the subsequent prediction of its mask.

This is why small-lesion segmentation is rarely just a boundary-drawing problem. When a lesion occupies only a tiny fraction of the image, searching the whole image is both costly and error-prone. The natural approach is to narrow the search space first, then segment within the narrowed region. Fully automatic segmentation methods, typified by U-Net and Attention U-Net [3], [4], try to learn both localization and segmentation implicitly within the same network, with no external help at all.

Another approach is to draw on external guidance about the lesion's location. A coarse bounding box around a suspicious region is one such cue, and is not merely an interactive convenience but a practical way to inject spatial knowledge into the model before detailed segmentation begins. Many prompt-based interactive segmentation methods have grown out of this idea, from simple operations such as clicks and scribbles [5], [6] to recent foundation models such as SAM and SAM-Med2D [1], [2].

We investigate a compact architecture in which a box prompt is supplied explicitly to multiple encoder and decoder stages. PGA-UNet represents the box as a Gaussian-smoothed plateau, modulates encoder features through PSG, and adds separately encoded prompt features to decoder skip attention through CAD. This design provides repeated, explicit access to the prompt within the network. Its motivation is to support segmentation when the target occupies a small or visually ambiguous region; preservation of lesion-specific internal features is not measured directly in this study.

The empirical evaluation compares the complete design with conventional ways of incorporating the same box, including a binary input channel and a prompt crop. Ablations examine the observed effects of the prompt representation and conditioning components. Because the conventional channel baseline and the full model use different prompt representations, their comparison evaluates the complete configurations. A binary-prompt PGA-UNet variant provides an additional comparison at a shared prompt representation.

The contributions of this work are as follows.

- We introduce PGA-UNet, a compact convolutional architecture that incorporates a Gaussian-smoothed box representation into encoder features through a Prompt Spatial Gate and into decoder skip attention through a Conditional Attention Decoder.
- We evaluate the architecture on BTXRD and FracAtlas using simulated lesion-covering boxes, conventional prompt baselines, and a comparison with SAM-Med2D at matched input resolution and scoring frame.
- We characterize the effects of the conditioning components, the observed response to a predefined prompt displacement, variability across four additional image-level splits for PGA-UNet, and computational cost in the tested environment.

Beyond these contributions, the system also exposes two support signals, computed directly from the trained model's output without ground-truth masks at inference: a self-assessment ranking heuristic, and a short candidate-region shortlist. Both are presented as preliminary and exploratory, and are not central contributions of this article.

II. RELATED WORK

For a long time, small-lesion segmentation in medical imaging has run into a simple fact: to segment a very small target well, a model must first narrow down where to look. In the fully automatic direction, this narrowing is learned implicitly inside the network. U-Net is the multi-scale encoder-decoder architecture that laid the foundation for biomedical segmentation [3], and Attention U-Net later added attention gates on the skip connections so the decoder could filter out irrelevant spatial responses more selectively [4]. Self-configuring pipelines such as nnU-Net go further still, automatically adapting preprocessing, network shape, and training schedule to each dataset [10]. These models raise automatic segmentation quality considerably, but still rest on a shared assumption: the network must localize and draw the mask from the input image alone, with no hint at all about where the lesion lies.

A different automatic strategy pairs detection with segmentation directly. Both source datasets used in this article report such a baseline in their own technical validation: BTXRD evaluates YOLOv8s-seg for automatic lesion detection and instance segmentation, and FracAtlas evaluates the same architecture for fracture regions [7], [8]. YOLOv8s-seg must localize the lesion itself, is scored under each dataset paper's own detection-style metric, and is not retrained or re-evaluated in this article; we cite it as literature context for how the two source datasets have been used for automatic segmentation, not as a directly comparable quantitative baseline under the box-prompted protocol used in the remainder of this article (Table 1).

In parallel, interactive medical segmentation has also developed, starting from the observation that even a small amount of user guidance can substantially simplify the problem. Early methods used clicks, scribbles, or geodesic cues [5], [6]; later, bounding boxes were added as a fast, coarse form of prompt. More recently, interactive models such as ScribblePrompt can handle several prompt types within a single framework, across many biomedical datasets [11], and transformer-based interactive segmenters such as SimpleClick extend this line of work to natural-image benchmarks with a plain vision-transformer backbone [17]. Broadly, these methods shift part of the localization burden from the model to the user. For very small X-ray lesions, that shift is especially valuable, since a clinician can usually point out the suspicious region coarsely even when an automatic system struggles to find it reliably in the whole image.

Promptable foundation models push this idea further. SAM pioneered turning prompt-driven segmentation into a general-purpose task [1]; SAM-Med2D and MedSAM brought it into medicine [2], [9]; adapter-based variants such as EMedSAM and Med-SA reduce the cost of specializing these models to medical data [12], [13]. These models are quite flexible, but they consume the prompt mainly at the mask decoder, and carry a very large backbone relative to the compact convolutional neural network (CNN) studied here.

Conditioning intermediate features on an external signal is also studied outside segmentation. FiLM applies a learned per-channel affine transformation to feature maps from a conditioning input [15], and SPADE conditions intermediate normalization layers on a spatial semantic layout for image synthesis [16]. PSG's per-channel multiplicative gate and CAD's decoder-level additive prompt term are architecturally closer to this general feature-conditioning family than to the input-concatenation baselines evaluated in this article; we do not claim feature-wise conditioning itself as a new principle, and instead study its application and empirical effect when applied to box-prompted CNN segmentation at multiple encoder and decoder stages together.

Our focus is a compact CNN with explicit box conditioning at encoder and decoder stages. The evaluated reference models differ in backbone, pretraining, prompt representation, and operating resolution. We therefore distinguish comparisons with a shared scoring frame from configuration-specific reference results and limit conclusions to the settings actually evaluated. In this article we study PGA-UNet directly against fine-tuned SAM-Med2D, MedSAM, and ScribblePrompt-UNet under this evaluation protocol. ScribblePrompt-UNet is itself a lightweight, prompt-receiving CNN with a parameter count close to PGA-UNet's (Table 16); this article compares the two empirically under the evaluated protocol but does not establish an architectural difference between PGA-UNet's

TABLE 1. Representative prior work, by prompt interface and architectural approach.

Work	Prompt interface	Architectural approach
U-Net [3]	None	Multi-scale skip-connected encoder-decoder
Attention U-Net [4]	None	Skip-connection attention gates, learned without prompt input
nnU-Net [10]	None	Self-configuring, dataset-adaptive automatic pipeline
YOLOv8s-seg [7], [8]	None (joint detect+seg)	One-stage detector-segmenter; the BTXRD/FracAtlas dataset papers' own automatic baseline, not retrained here
DeepIGeoS and related [6]	Clicks, scribbles, geodesic cues	Interaction-aware segmentation network
ScribblePrompt [11]	Scribbles, clicks, boxes	Single lightweight UNet trained across many biomedical datasets and prompt types
SimpleClick [17]	Clicks	Plain vision-transformer backbone for interactive segmentation
SAM-Med2D [2]	Boxes, points, or masks	Large pretrained transformer backbone; prompt consumed mainly at the mask decoder
MedSAM [9]	Boxes (box-only by design)	Large pretrained transformer backbone; prompt consumed mainly at the mask decoder
EMedSAM / Med-SA [12], [13]	Same as SAM	Lightweight adapters on a frozen SAM backbone
This work	Box	Compact CNN; Gaussian plateau prompt injected at encoder (PSG) and decoder (CAD)

PSG/CAD mechanism and how ScribblePrompt-UNet's own network internally consumes a prompt, since that comparison is outside the scope of what we verified here.

III. METHOD

A. OFFLINE AND ONLINE FRAMEWORK

The system operates in two stages. In the offline stage, the model is trained on X-ray images that already have lesion polygons. Each polygon is turned into a training prompt by generating a simulated bounding box, then converting that box into a Gaussian-smoothed plateau heatmap. The network learns to map the (image, prompt) pair to a binary lesion mask. In the online stage, the user only needs to draw a coarse bounding box around the suspicious region on a new image; the same prompt-construction procedure is applied again, and the trained model produces the final predicted mask. Only the plateau-heatmap encoding (rasterize, Gaussian-smooth, resize-and-pad, Section III-C) is reapplied to a clinician-drawn box; the $3.0\times$ center-scaling and off-center shift described there are simulation rules used to turn a ground-truth polygon into a box for training and evaluation, not a transformation applied to a box someone actually draws.

What sets this apart is where the prompt lives: instead of entering only once at the first layer, it is kept as a dense spatial prior, active throughout the encoder and decoder, which should help when the box is coarse, slightly off-center, or wrapped tightly around a very small lesion.

An interactive demonstration of the online stage is given in Figure 1: the user draws a box on the X-ray image, and the model returns the mask together with CAD's prompt-use score and the self-assessment score of Section III-F, both of which require no ground truth.

B. PROBLEM SETTING

Given a grayscale radiograph $\mathbf{I} \in \mathbb{R}^{H\times W}$ and a user-provided bounding box $\mathbf{B}$, the goal is to predict a binary lesion mask $\hat{\mathbf{M}}$. Let $\mathbf{H}(\mathbf{B})$ be the prompt heatmap derived from the box. PGA-UNet predicts

$$\hat{\mathbf{M}} = \mathbf{1}\left[f_{\text{PGA}}\left(\mathbf{I}, \mathbf{H}(\mathbf{B}); \theta\right) \geq 0.5\right]. \tag{1}$$

C. PROMPT REPRESENTATION

Instead of encoding the input prompt as a hard binary rectangle, we rasterize the interior of the box into a plateau map at the original image resolution, soften its edges with a fixed 31×31 Gaussian kernel (OpenCV's automatic σ for this kernel size, $\sigma \approx 5.0$ px), and only then resize and pad this heatmap down to the target square resolution together with the image. The kernel size does not change with the target resolution (128×128, 256×256, or 512×512): applying it once, at the original resolution, before downsampling, keeps the effective blur consistent regardless of which square frame the network ultimately sees. The plateau covers exactly the area of the box (expanded if applicable), with no additional minimum margin added; coverage of the region of interest is already guaranteed by the box-expansion step, so no separate margin step is needed.

At evaluation time, the prompt is tested under two fixed conditions. The covering condition (`center_zoom`) scales the tight lesion box outward from its own center by a fixed factor of 3.0. The off-center condition (`center_shift`) applies the same scaling, then shifts the box by a fraction of the tight-box size (capped by `shift_ratio`=0.5), expanding it back if needed so the box still fully contains the lesion. At test time, this shift is fixed per lesion (seeded from the sample index), so every model is evaluated on exactly the same set of off-center boxes; during training, the shift is redrawn randomly at each iteration. For each dataset and resolution, we train a single model using `center_mixed`: `center_shift` is chosen with probability 0.8 and `center_zoom` with probability 0.2, while at test time the two conditions are scored separately. The task scope assumes the user has already identified the suspicious region and deliberately draws a box that covers the lesion; partial-coverage boxes, negative boxes, and off-target boxes all lie outside this controlled protocol, and whether a clinician can find the correct region in the first place is a separate question, outside the scope of this article.

Let (x_1, y_1, x_2, y_2) denote the tight lesion box, with $w = x_2 - x_1$ and $h = y_2 - y_1$. The simulated covering box uses exactly the fixed center-scaling rule above, applied during both training and evaluation. The off-center box reuses that scaled box, then applies the deterministic test-time shift rule implemented in `dataset.py`, while still preserving lesion coverage. The scale factor, shift ratio, and Gaussian kernel size were all chosen from preliminary experiments on the

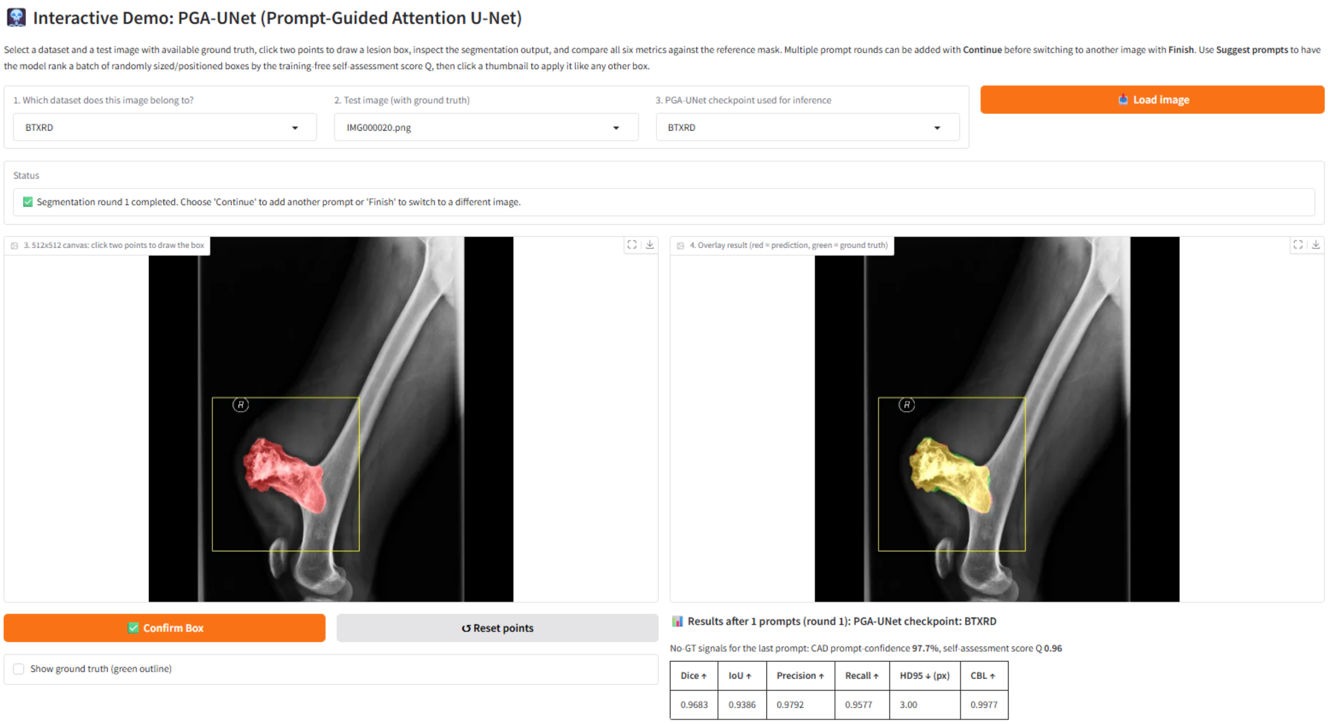

FIGURE 1. Interactive demonstration of the online stage on a BTXRD case. A user draws a box on the left canvas; the right panel overlays the predicted mask, and the panel below reports the reference metrics together with the no-ground-truth CAD prompt-use score and the self-assessment score *Q*. A FracAtlas case is shown in the same layout in the supplementary material.

training and validation data and then locked; the test split played no part whatsoever in choosing the protocol or any hyperparameter. These are simply fixed protocol values used throughout the article, not a globally optimal configuration. This simulation naturally cannot substitute for a prompt actually drawn by a radiologist, but it offers a controlled way to measure the model's sensitivity to a predefined prompt displacement.

In summary, prompt construction consists of three steps: (1) obtain a box, either the annotated polygon's tight box expanded under the covering or off-center simulation rule above (training and evaluation) or a box a user draws directly online with no further expansion (Section III-A); (2) rasterize that box into a plateau mask at the original image resolution; (3) smooth the plateau's edge with the fixed Gaussian kernel before resizing to the target resolution. This avoids the hard-edge dependence of a binary box while still keeping a clear spatial prior.

D. PROMPT SPATIAL GATE

Let $\mathbf{x}^l$ denote the encoder feature map at level l. PSG modulates it with a prompt-derived attention map $\mathbf{A}^l_{\mathrm{PSG}}$:

$$\tilde{\mathbf{x}}^l = \mathbf{x}^l \odot \left(1 + \alpha^l_{\mathrm{PSG}} \mathbf{A}^l_{\mathrm{PSG}}\right), \tag{2}$$

where $\odot$ denotes element-wise multiplication, and α^l_{PSG} is a learnable parameter, initialized at 0.1 and clamped to $[0, 1]$. The residual form preserves the original feature stream while emphasizing prompt-relevant regions. $\mathbf{A}^l_{\mathrm{PSG}}$ is produced from the prompt heatmap (resized by bilinear interpolation to $\mathbf{x}^l$'s spatial size when needed) by a learned 1×1 convolution followed by a sigmoid, projecting it to the same number of channels as $\mathbf{x}^l$; it is therefore a per-channel gate at each spatial location, not a single map shared identically across channels.

E. CONDITIONAL ATTENTION DECODER

CAD extends decoder attention by conditioning the skip gate on prompt-encoded features. Let $\mathbf{g}^l$ be the decoder gating feature and $\mathbf{p}^l_{\mathrm{enc}}$ the prompt encoding at the same scale. The conditioned gate is

$$\mathbf{g}'^{\,l} = \mathbf{g}^l + c^l \alpha^l_{\mathrm{CAD}} w_l \mathbf{p}^l_{\mathrm{enc}}, \tag{3}$$

where c^l is a prompt-use coefficient derived from the decoder, α^l_{CAD} is learnable, and w_l is a fixed depth coefficient. This construction supplies a prompt-dependent term to decoder attention; its observed response to the predefined box displacement is reported in the prompt-sensitivity analysis (Section V-F).

Equation (3) specifies how the prompt-conditioned gating signal $\mathbf{g}'^{\,l}$ is formed; it does not by itself specify the rest of the decoder block. After $\mathbf{g}'^{\,l}$ is computed, it is used as the gating input to the same grid-attention mechanism as the automatic Attention U-Net baseline, attending over the skip feature at that level; this is the sense in which CAD extends, rather than replaces, that gate. The block then adds two further terms not shown in Equation (3): a fixed residual bypass of 0.3 times the pre-attention skip feature around the attention output, and a learnable-weighted term (initialized at 0.05) adding the resized prompt heatmap directly to the block's output. We report these for reproducibility; their individual contribution is not separately ablated in this article.

The prompt encoder inside CAD consists of two 3×3 convolutional layers with instance normalization and ReLU. The prompt-use coefficient c^l is produced by global average pooling, followed by a 1×1 convolution and a sigmoid; this coefficient is computed only inside the gate and used to scale the prompt term at that level; it is per-decoder-level (four values), and we do not analyze it quantitatively as a separate model output in this article's results. The interactive demo (Figure 1) displays the mean of these four CAD mixing coefficients as a descriptive internal quantity; we do not interpret it as a calibrated measure of prompt reliance or prediction correctness. The decoder's depth coefficients are fixed at $\{1.0, 0.7, 0.4, 0.2\}$ from deep to shallow. With a feature scale of 4, the channel widths are $\{16, 32, 64, 128, 256\}$ from the shallowest encoder stage to the bottleneck, and the whole model has about 2.95 million parameters. These protocol values were all chosen from preliminary experiments on the training and validation sets, held fixed throughout the main evaluation, with the test split never involved in tuning; we make no claim of optimality. The CAD mixing coefficient is parameterized through a sigmoid and initialized near 0.3.

PSG applies a prompt-derived multiplicative factor at each encoder level. Since both the gain α^l_{PSG} and the attention map $\mathbf{A}^l_{\mathrm{PSG}}$ are non-negative, this factor is always at least one: locations with a zero attention value are unchanged, while locations with a positive value are amplified according to the learned gain, and the amount of amplification can differ by level rather than being fixed once at the input. CAD introduces a separately encoded prompt term, $\mathbf{p}^l_{\mathrm{enc}}$, into decoder skip attention at each level, through a coefficient c^l recomputed at that level on every forward pass. The distinction from input-channel conditioning, such as the U-Net with a binary prompt channel baseline (Section IV), is the location and explicit form of prompt injection: that baseline folds the prompt into the input once, after which it is not distinguished from image content. These operations specify the architecture; whether this difference also produces the measured gap over the prompt-channel baseline (Tables 4 and 13) is assessed through the reported experiments, not proved from the equations alone.

In summary, the method consists of four steps: (1) preprocess the image and prompt to the same square resolution; (2) encode the prompt-relevant spatial prior with PSG in the encoder; (3) propagate prompt-conditioned context through CAD in the decoder; (4) predict the final binary lesion mask.

F. TRAINING-FREE SELF-ASSESSMENT SCORE

Besides the predicted mask, PGA-UNet also returns a scalar Q, an exploratory ranking score computed from prediction decisiveness and agreement under prompt perturbations. It does not require ground-truth masks at inference; it is computed only from the model's own output at inference time, as the average of two cues. Its association with actual segmentation quality is evaluated separately using the annotated test data (Section V-M). Q is used as a support signal, letting the user look more critically at a given mask, not as a core component of the method.

The first cue, S_{sharp}, measures how decisive the predicted mask is. Let p be the sigmoid probability map and $\Omega = \{p > 0.5\}$ the predicted positive region. Then

$$S_{\mathrm{sharp}} = \mathrm{clip}_{[0,1]}\left(\frac{1}{|\Omega|}\sum_{u\in\Omega}\big(2p(u) - 1\big)\right). \tag{4}$$

A mask with high confidence, i.e., probabilities close to 1, scores close to 1; conversely, a mask with many pixels sitting near the decision threshold scores lower. If $\Omega = \emptyset$ (the model predicts no positive pixel at all), S_{sharp} is defined as 0 rather than left undefined.

The second cue, S_{stab}, measures how stable the prediction is under mild prompt perturbation. The box is randomly nudged K times, each time with a small shift and scale change; the model is rerun for each nudged box, and S_{stab} is exactly the average intersection-over-union (IoU) between the original binary mask $\hat{\mathbf{M}}$ and the masks obtained from those nudges, $\hat{\mathbf{M}}_k$:

$$S_{\mathrm{stab}} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{IoU}\Big(\hat{\mathbf{M}}, \hat{\mathbf{M}}_k\Big). \tag{5}$$

A prediction that stays nearly unchanged when the box is nudged slightly is considered more trustworthy than one that collapses or drifts elsewhere. For this heuristic, an empty-empty mask pair contributes zero, so consistently empty predictions do not receive a high score from this term.

The final score is $Q = \frac{1}{2}\big(S_{\mathrm{sharp}} + S_{\mathrm{stab}}\big)$, with $K = 6$, a maximum shift of 0.12 of the box side, and a scale factor within ± 0.10. It should be emphasized that Q is only a heuristic ranking signal, not a calibrated probability: a value of 0.7 does not mean a 70% probability that the mask is correct. Q also presupposes that a lesion lies inside the box: it measures mask quality, not lesion presence.

We reuse this same score to shortlist candidate regions when no prompt is given: a set of boxes with varying size and position is sampled, each box is scored with Q, and the highest-scoring few are returned for the clinician to accept, adjust, or discard. This is only a scaffold for a small suggestion feature we are working toward, not an automatic detector.

IV. EXPERIMENTAL SETUP

A. DATASETS

We run our experiments on two public bone X-ray datasets. BTXRD is a primary bone tumor dataset with polygon annotations supporting classification, localization, and segmentation [7]. FracAtlas is a fracture dataset with the same label types [8]; we convert its segmentation annotations to the same polygon format used for BTXRD before use, and the conversion is released with the reproducibility package. Both

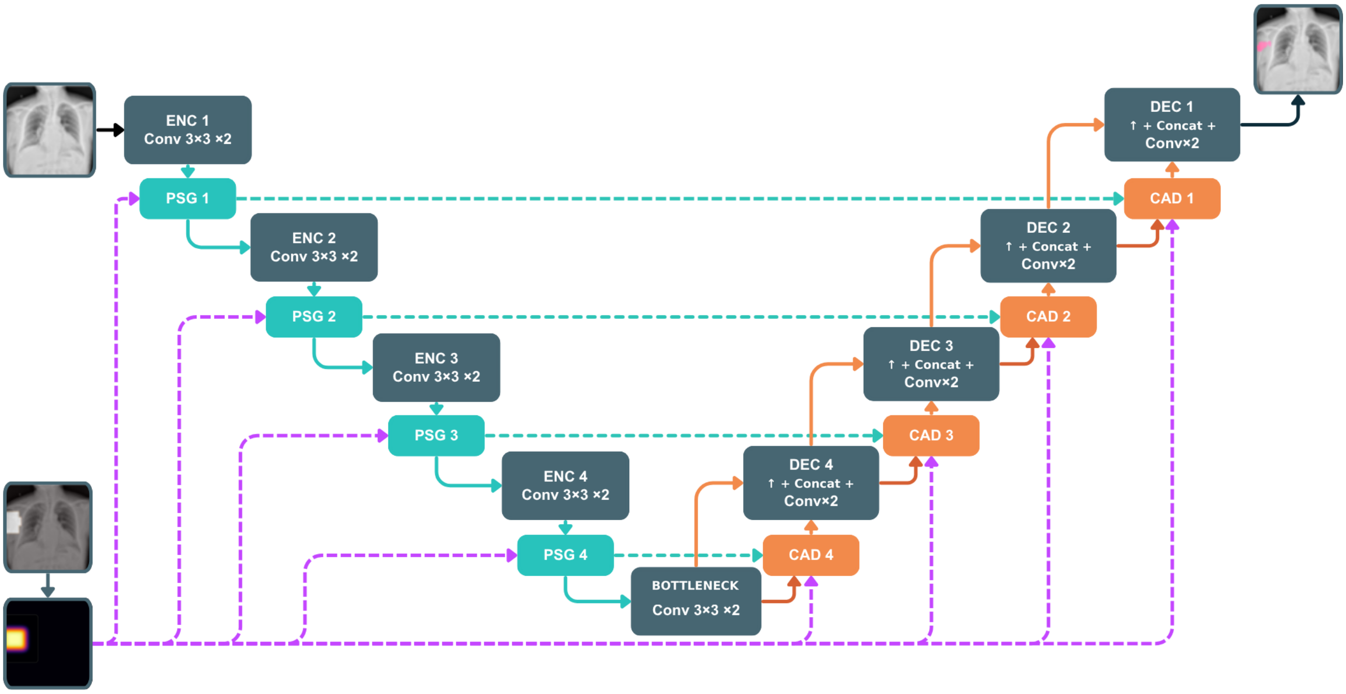

FIGURE 2. **Overview of PGA-UNet. Prompt information derived from the bounding box is injected into encoder features through PSG and reused in decoder skip attention through CAD.**

TABLE 2. **Image-level train / validation / test splits. Each lesion polygon is one promptable sample.**

	Images			Lesion polygons		
Dataset	Train	Val	Test	Train	Val	Test
BTXRD	1,493	187	187	1,848	238	232
FracAtlas	573	72	72	730	95	92

datasets are split at the image level, so that every polygon belonging to the same image always falls in the same split (Table 2).

Table 2 contains 917 converted lesion polygons for FracAtlas (730 + 95 + 92), whereas the original FracAtlas release reports 922 fracture instances. These totals have not been reconciled at the annotation-ID level. The reported segmentation results therefore refer to the converted annotation set summarized in Table 2. Because the public metadata do not provide enough information to verify strict patient-level separation, images from the same patient could in principle still fall into different splits; we regard this as a limitation of this evaluation. In image-level evaluation, if an image has multiple lesions, the model is run separately for each box, and the resulting probability maps are merged by a pixelwise maximum before thresholding at 0.5; the ground-truth masks are likewise merged by union.

Input images are resized isotropically to preserve aspect ratio, then padded to a square resolution of 128×128, 256×256, or 512×512, depending on the configuration. Images, masks, and prompt maps all undergo this same geometric preprocessing.

B. TRAINING DETAILS

All CNN models were implemented in PyTorch and trained on a single NVIDIA Tesla T4 GPU with 16 GB memory. PGA-UNet and Attention U-Net were trained separately for each dataset and each resolution; in other words, this article studies one shared architecture family instantiated as separate models, with separate parameter sets for BTXRD and FracAtlas, rather than training a single joint model on both. We used the AdamW optimizer with learning rate 10^{-4} and weight decay 10^{-4}; the loss was the sum of binary cross-entropy and Dice loss. Because the segmentation targets are often small, we additionally tested training PGA-UNet at 512×512 with the Dice term replaced by a size-conditioned Tversky loss, and in a separate run, by a Focal Dice loss, with every other setting held fixed. These two replacements are mutually exclusive and are reported later only as a negative control on the training objective. Batch size was 4, training ran for at most 150 epochs, with early stopping at patience 15. All main and ablation experiments share the same fixed training seed, 22120196, which seeds Python, NumPy, and PyTorch on both CPU and CUDA. The primary model-selection criterion was image-level merged Dice, measured on the validation set under the `center_shift` condition.

During training, we applied augmentation consisting of horizontal flipping and random rotation within $\pm 15°$. PGA-UNet is trained with `center_mixed`: `center_shift` is chosen with probability 0.8 and `center_zoom` with probability 0.2. Because the prompts are all simulated from labeled lesion regions, this is a prompted-segmentation protocol under controlled conditions; the article should be read accordingly, namely as a study of simulated lesion-covering boxes around user-identified lesions, not as an evaluation of unconstrained lesion detection or of boxes fully drawn by a clinician in real practice.

For the fine-tuned SAM-Med2D comparison, the pre-trained image encoder was frozen, with only its lightweight Adapter layers trained further, while the prompt encoder and mask decoder were fully updated on each dataset, under the same split protocol. SAM-Med2D was fine-tuned with box prompts only, using the same center-scaled covering and off-center protocol as PGA-UNet; the point-prompt branch and iterative point refinement were both disabled. This box construction differs from the small random-jitter `get_boxes_from_mask` sampling used by the original SAM-Med2D authors, which we did not apply here. Both validation during SAM-Med2D fine-tuning and the final test-split evaluation report both prompt conditions separately; batch size was 4, at most 150 epochs, early stopping patience 30, and learning rate 10^{-5}. This matched protocol reduces differences caused by prompt distribution and resolution, but cannot eliminate differences in model scale, initialization, or optimization.

We additionally ran stability experiments using Monte Carlo cross-validation, that is, repeated random image-level splits, as opposed to strict non-overlapping k-fold partitioning. These experiments keep the training seed fixed at 22120196 and vary only the split seed, across the values 1, 2, 3, and 4.

Two details are worth noting when interpreting the evaluation results. First, prompt construction itself does not depend on resolution: the plateau heatmap is built and smoothed with a fixed 31×31 Gaussian kernel at the original image resolution, and only then resized and padded down to 128×128, 256×256, or 512×512 together with the image. In other words, the same absolute blur is compressed differently at each target resolution, rather than being parameterized separately for each. Second, the Monte Carlo experiments are only auxiliary stability evidence, not intended to replace the fixed train/validation/test split used throughout the main results tables.

C. BASELINES AND METRICS

We compare PGA-UNet against an automatic baseline, Attention U-Net [4], and three fine-tuned promptable baselines: SAM-Med2D [2], MedSAM [9], and ScribblePrompt-UNet [11]. Attention U-Net is trained without a prompt, to show how hard the task becomes when localization must be fully automatic. Among the promptable baselines, SAM-Med2D is fine-tuned and evaluated at 256×256 using the same dataset splits and the same box prompts as PGA-UNet, so PGA-UNet at 256 against SAM-Med2D at 256 is the article's resolution-matched prompt comparison. MedSAM and ScribblePrompt-UNet run at their own native resolutions (1024 and 128); Table 7 reports these alongside PGA-UNet's own 128 and 512 results as configuration-specific reference results, each scored in its own frame, and does not use them to establish a cross-model ranking.

To compare the proposed design with conventional ways of using the same box, we evaluate two additional baselines based on the Attention U-Net backbone, without PGA-UNet's Gaussian-prior heatmap, Prompt Spatial Gate, or Conditional Attention Decoder. The channel baseline receives a binary box mask as an additional input channel, whereas the crop baseline receives the image region defined by the box. Only the crop baseline retains the backbone's original attention-gated skip connections unmodified, since the box only changes what image region enters an otherwise-standard Attention U-Net, so we refer to it as Attention U-Net + prompt crop; the channel baseline instead concatenates the mask at the input and decodes through plain, non-gated skip connections, so it is not attention-gated in the same sense as the automatic Attention U-Net reference, and we name it accordingly, U-Net + binary prompt channel. The full PGA-UNet model uses a Gaussian-smoothed heatmap together with explicit encoder and decoder conditioning. These comparisons therefore concern the complete tested configurations, combining differences in prompt representation, backbone gating, and PGA-UNet's own conditioning mechanism, not any one of these in isolation. Both baselines are trained under the same covering and off-center box protocol as PGA-UNet. The binary-prompt PGA-UNet ablation (Section V-G) provides a comparison using the same binary box representation as the channel baseline, without isolating every architectural component. When comparing models against each other, we use the pasted-back full-frame metric for the prompt-crop baseline, not the diagnostic value measured inside the crop frame.

Every comparison table in this article reports Dice and IoU for region overlap, CBL (Center-Based Localization) for localization, and HD95 (95th-percentile Hausdorff Distance) for boundary distance. Dice and IoU are fairly strongly correlated overlap measures; we place them side by side only for the reader's convenience in cross-checking. On the small-lesion subsets specifically, IoU has a low absolute value and is quite sensitive to a few boundary pixels, so Dice is the primary overlap metric used there. Let (x_p, y_p) be the predicted mask centroid, (x_g, y_g) the ground-truth centroid, and D_{GT} the diagonal length of the ground-truth bounding box; CBL is defined as follows:

$$\mathrm{CBL} = \max\left(0, 1 - \frac{\sqrt{(x_p - x_g)^2 + (y_p - y_g)^2}}{D_{\mathrm{GT}}}\right). \quad (6)$$

If the ground-truth mask is empty, the sample is excluded from the CBL average entirely; if the ground-truth is non-empty but the predicted mask is empty, CBL is set to 0 for that sample. HD95 is computed from bidirectional boundary distances. When both masks are empty, HD95 is set to 0; if only one mask is empty, HD95 is set to the side length S of whichever frame the sample is scored in (128, 256, or 512 pixels for a native-frame table; the shared frame's own side length, e.g. 512 in Table 8, for a shared-frame table, applied identically to every model scored there). No sample is discarded from the computation. For comparisons spanning resolutions, we additionally report HD95 in normalized form, $\mathrm{HD95}/(\sqrt{2}\,S)$, since a raw pixel distance cannot be directly compared across the 128, 256, and 512 frames. Some subgroup comparisons mix models trained at 256×256 and 512×512; in that case, everything is scored together in a shared 512×512 frame, with the 256×256 predictions upsampled before scoring, so every model is measured against exactly the same ground-truth mask, rather than in its own native frame. Cross-model comparisons used to support the main conclusions are restricted to a shared scoring frame: the conventional prompt baselines and PGA-UNet at 512×512 (Table 11), and SAM-Med2D and PGA-UNet at 256×256 (Table 6). The lower-lesion-area image subset comparison with SAM-Med2D (Table 8) uses 256×256 model inputs and a shared 512×512 scoring frame. Other operating configurations, including PGA-UNet's own 128 and 512 results and the MedSAM and ScribblePrompt-UNet results, are documented separately in Table 7 as configuration-specific reference results, each scored in its own frame (PGA-UNet at 128 or 512; MedSAM and ScribblePrompt-UNet at the original image resolution, not at the 128 or 1024 frame their own network internally resizes to); their raw HD95 values are shown for completeness but are left unbolded. Dice, IoU, and CBL are dimensionless, but they need not remain unchanged after mask resampling and rasterization: a model run at a lower native resolution discretizes a lesion's boundary into fewer pixels before any of these metrics is computed. Consequently, results scored in different frames are not used to establish a cross-model ranking in this article. Normalizing HD95 changes its distance scale but does not remove differences introduced by prediction and ground-truth resampling. Table 15 additionally reports precision and recall for the image-level merged mask; these two metrics are not used in any other comparison in this article, and recall there is this standard pixel-level quantity, not a lesion-level detection recall across a cohort of images, which this article does not measure.

For the training-free self-assessment score, each lesion prompt yields a score Q together with the true thresholded Dice of its own predicted mask; we report the Pearson and Spearman correlation between the two, separately for the covering and off-center conditions. For the candidate-region shortlist, each test image is sampled with 50 boxes whose side length is uniform in $[0.15, 0.55]$ of the frame, each box is scored by Q, and the five highest-scoring boxes are kept; each retained box is then scored by coverage, the fraction of ground-truth lesion pixels it contains, and no predicted mask enters this coverage measure. From the same 50-box pool we also compute a random-5 baseline (mean over 2,000 resamples per image, without replacement) and an oracle-5 upper bound (the 5 boxes with the truly highest coverage), to check whether Q-ranking adds value over chance. As a negative control, the same 50-box sampling is also run on the lesion-free images of each dataset (1879 for BTXRD, 3366 for FracAtlas); with no lesion present there is no coverage to compute, so we report only the distribution of Q.

D. STATISTICAL ANALYSIS

The main baseline comparisons (Tables 4, 5, 6) report descriptive point estimates on the fixed test splits: each configuration in these tables is a single fixed-run comparison, one trained checkpoint per configuration rather than repeated training runs, and no paired uncertainty estimates are reported for these comparisons. The existing ablation analysis (Section V-G) uses paired per-image Dice values with a two-sided sign-flip randomization test and percentile bootstrap confidence intervals, each based on 20,000 resamples, with Holm correction as specified in Table 13. These results are conditional on the evaluated split and trained checkpoints. Four additional image-level splits (Section V-H) characterize variation in PGA-UNet's own performance while keeping its training seed fixed; they do not estimate variation due to the training seed itself, nor do they establish the stability of differences from competing models. A separate training-seed stability analysis (Section V-I), restricted to PGA-UNet on the original fixed split, reports an interim result from $n = 2$ of the four planned training seeds at the time of this draft, with the remaining two seeds still training; like the Monte Carlo splits, it does not by itself establish the stability of any baseline comparison. The exploratory self-assessment correlation analysis (Section V-M) retains its own permutation p-values and bootstrap 95% intervals for Spearman's ρ. All resampling uses the fixed seed 22120196.

E. SCOPE OF ANALYSIS

This article focuses only on prompt-guided segmentation itself. A lesion-screening classification stage was explored in a separate study and is not part of this article. The self-assessment score and the candidate-region shortlist are treated only as auxiliary analyses, within the same narrow scope already stated in Section III-F; apart from one negative control on lesion-free images, they are not extended any further here. As a further exploratory check on that same weakness, Section V-M also reports results from a separate, external post-hoc gate (MedSigLIP); that check is BTXRD-only, does not include a post-gate Dice or other segmentation-quality evaluation, and is not a core contribution of this article. To keep the article compact, we retain only the subgroup analyses that most directly support the main claim: the role of prompt-guided localization (Section V-C), sensitivity to controlled prompt displacement as defined above (Section V-F), and observed results on the images with the lowest total lesion area (Section V-E, where the subgroup definition and its coverage of each test split are given). It should also be reiterated: partial-coverage boxes, negative boxes, and off-

TABLE 3. PGA-UNet across resolutions, image-level merged. Dice, IoU, CBL and HD95 (native px and normalized HD95/($\sqrt{2}\,S$)) given as covering / off-center.

Dataset	Res.	Dice	IoU	$\mathrm{HD95_{px}}$	$\mathrm{HD95}_n$	CBL
BTXRD	128	0.714 / 0.717	0.572 / 0.577	5.8 / 5.3	0.032 / 0.029	0.859 / 0.857
	256	0.762 / 0.743	0.633 / 0.611	10.9 / 12.8	0.030 / 0.035	0.902 / 0.885
	512	**0.782 / 0.774**	**0.661 / 0.652**	22.8 / 24.5	0.031 / 0.034	**0.918 / 0.912**
FracAtlas	128	0.596 / 0.542	0.445 / 0.389	8.1 / 9.5	0.044 / 0.053	0.849 / 0.790
	256	0.629 / 0.594	0.470 / 0.436	10.6 / 11.7	0.029 / 0.032	0.895 / 0.849
	512	**0.727 / 0.713**	**0.585 / 0.570**	10.5 / 10.4	**0.015 / 0.014**	**0.913 / 0.897**

target boxes are outside the scope of this study, so the article's claims do not rely on them.

V. RESULTS

Unless stated otherwise, every table reports current-protocol numbers: `center_mixed` training with 80% `center_shift` and 20% `center_zoom`, 150 epochs, model selection by image-level merged validation Dice under `center_shift`, and image-level merged test evaluation. The two test prompt conditions, covering (`center_zoom`) and off-center (`center_shift`), are reported separately whenever a table carries both.

A. MAIN RESULTS ACROSS DATASETS AND RESOLUTIONS

PGA-UNet was trained from scratch at 128×128, 256×256, and 512×512 on each dataset, giving six models. Table 3 reports them under both prompt conditions.

The reported Dice values increase from the 128 to the 256 and 512 configurations on both datasets and under both prompt conditions, with FracAtlas gaining more. Each configuration is scored in its own output frame, so these are operating-configuration results that combine changes in model input resolution and mask sampling; they do not isolate an internal feature-preservation mechanism or provide a common-frame estimate of the effect of input resolution alone. Because HD95 in raw pixels cannot be compared across differently sized frames, we also report the normalized form: it stays near 0.03 across the BTXRD resolutions and decreases on FracAtlas as the frame grows. Note that each model is evaluated only on the dataset it was trained on; the article makes no claim about cross-dataset transfer.

B. COMPARISON AGAINST AUTOMATIC AND PROMPT-MATCHED BASELINES

With no support at all in finding the lesion, Attention U-Net reaches only 0.517 Dice on BTXRD and 0.342 on FracAtlas at 512×512, with HD95 exceeding 120 pixels on both datasets (Table 4). The no-prompt reference addresses a different input setting than every other row in this table, and is not used to attribute the differences below to the proposed architecture.

Under covering prompts at 512×512, PGA-UNet obtains Dice scores of 0.782 on BTXRD and 0.727 on FracAtlas, compared with 0.760 and 0.593 for the binary prompt-channel baseline and 0.741 and 0.590 for the prompt-crop baseline (Table 4). An additional comparison is available

TABLE 4. Comparison at 512 $\times$ 512 under covering prompts. Attention U-Net (no prompt) receives no box; the other rows use the same box PGA-UNet receives. The Delta Dice column is each row minus PGA-UNet.

Dataset	Model	Dice	ΔDice	IoU	HD95	CBL
BTXRD	Attention U-Net (no prompt)	0.517	-0.265	0.418	120.4	0.620
	U-Net + binary prompt channel	0.760	-0.022	0.635	33.0	0.895
	Attention U-Net + prompt crop	0.741	-0.041	0.615	24.7	0.912
	PGA-UNet	**0.782**	—	**0.661**	**22.8**	**0.918**
FracAtlas	Attention U-Net (no prompt)	0.342	-0.385	0.253	139.9	0.478
	U-Net + binary prompt channel	0.593	-0.134	0.434	22.9	0.887
	Attention U-Net + prompt crop	0.590	-0.137	0.451	22.0	0.865
	PGA-UNet	**0.727**	—	**0.585**	**10.5**	**0.913**

from the binary-prompt PGA-UNet ablation (Table 12): it obtains 0.788 and 0.668 respectively, using the same binary box representation as the channel baseline. This provides descriptive support for the explicit conditioning design beyond a change to Gaussian prompt encoding. These baselines control access to the box location, but the full PGA-UNet model also differs in prompt representation from the channel and crop baselines, so the comparisons here do not separately identify the effects of every architectural component, and no paired uncertainty estimates are reported for these baseline differences.

In the reported point estimates, the differences between the no-prompt and conventional prompted configurations are larger than the differences among the prompted configurations. The latter comparisons, between PGA-UNet and the two conventional prompt baselines, provide the more specific evidence for the proposed conditioning design.

C. ATTENTION U-NET TOP-DICE AND BOTTOM-DICE SUBSETS

To probe the same comparison on a different split of the test images, we ranked them by Attention U-Net Dice, kept the top 50 and bottom 50 per dataset, and re-ran all four models on those exact images (Table 5, covering prompts). On the BTXRD top-50 subset, the three prompted configurations have Dice values between 0.844 and 0.862. On the FracAtlas top-50 subset, their values span a wider range, from 0.605 to 0.754. On the bottom-50, Attention U-Net's Dice is far lower, down to 0.031 on BTXRD and 0.169 on FracAtlas, while PGA-UNet's is 0.712 and 0.687, and PGA-UNet leads the two conventional prompt baselines by 0.04 to 0.07 Dice on BTXRD and by 0.10 to 0.19 on FracAtlas within this group.

These subsets describe the selected images and are not independent evidence that the proposed method preferentially benefits intrinsically difficult cases: the Attention U-Net top- and bottom-Dice groups are defined entirely by that same baseline's own predictions, not by any external difficulty criterion, and on FracAtlas, whose test split has only 72 images (Table 2), the two groups are not independent samples: by construction at least $50 + 50 - 72 = 28$ images must belong to both, so the two rows are less a comparison between two disjoint image sets than a comparison between the 22 images each group excludes; on BTXRD, with 187 test images, the two groups share no images. We report the FracAtlas split as computed, but this overlap should be kept in mind when

TABLE 5. Top-Dice and bottom-Dice subsets at 512 × 512, 50 images per group, covering prompts. Groups are defined by plain Attention U-Net.

Dataset	Group	Model	Dice	IoU	HD95	CBL
BTXRD	Top-50	Attention U-Net	0.872	0.775	35.0	0.945
		U-Net + channel	0.846	0.740	24.3	0.942
		AttUNet + crop	0.844	0.739	27.5	0.946
		PGA-UNet	0.862	0.763	19.2	0.944
	Bottom-50	Attention U-Net	0.031	0.016	277.8	0.105
		U-Net + channel	0.668	0.541	43.3	0.831
		AttUNet + crop	0.652	0.518	17.6	0.875
		PGA-UNet	**0.712**	**0.589**	23.2	**0.879**
FracAtlas	Top-50	Attention U-Net	0.493	0.365	85.7	0.640
		U-Net + channel	0.605	0.447	26.6	0.887
		AttUNet + crop	0.658	0.515	24.9	0.881
		PGA-UNet	**0.754**	**0.615**	10.6	**0.927**
	Bottom-50	Attention U-Net	0.169	0.106	182.3	0.315
		U-Net + channel	0.574	0.416	25.3	0.875
		AttUNet + crop	0.509	0.369	26.0	0.834
		PGA-UNet	**0.687**	**0.535**	11.8	**0.896**

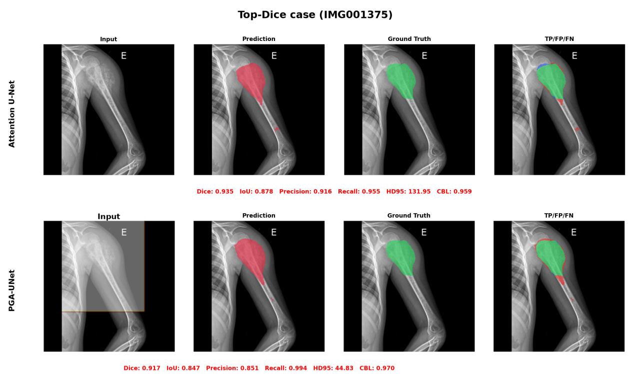

FIGURE 3. Attention U-Net (no prompt) vs PGA-UNet on BTXRD, Attention U-Net top-Dice case (IMG001375). Both models do well on an easy case.

reading its top-50/bottom-50 gap.

D. COMPARISON AGAINST PROMPTABLE BASELINES

The evaluated promptable references are SAM-Med2D, MedSAM, and ScribblePrompt-UNet. Tables 6 and 7 report their zero-shot and fine-tuned results under the specified box-prompt protocol. Within each evaluated model configuration, fine-tuning yields higher observed Dice than zero-shot initialization on both datasets. Comparisons used to support the main conclusions are restricted to the shared scoring frames described below.

At a shared input and scoring resolution of 256×256, PGA-UNet obtains Dice scores of 0.762 and 0.629 on BTXRD and FracAtlas, compared with 0.630 and 0.563 for the evaluated fine-tuned SAM-Med2D configuration (Table 6). The boundary results are mixed: PGA-UNet has lower HD95 on BTXRD (10.9 vs 17.3), whereas SAM-Med2D has lower HD95 on FracAtlas (8.0 vs 10.6). These fixed-split results concern the tested configurations and do not establish a general ranking of CNN and foundation-model architectures.

The MedSAM and ScribblePrompt-UNet results, together with PGA-UNet's own 128 and 512 configurations, are retained as configuration-specific reference results in Table 7

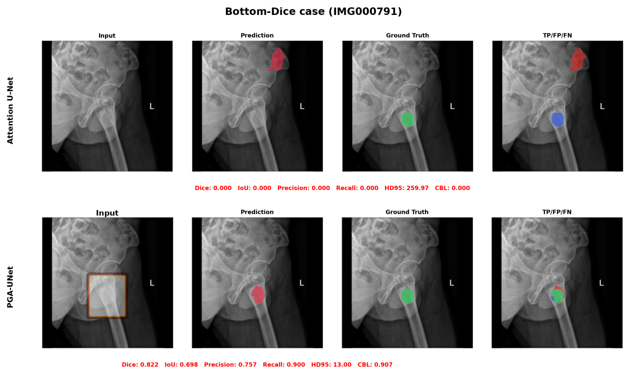

FIGURE 4. Attention U-Net (no prompt) vs PGA-UNet on BTXRD, Attention U-Net bottom-Dice case (IMG000791). Without a prompt, Attention U-Net's prediction is badly mislocated; the prompt-conditioned PGA-UNet still recovers the lesion.

TABLE 6. Comparison with SAM-Med2D using 256 × 256 model inputs and a shared 256 × 256 scoring frame, under covering prompts, image-level merged. Values are descriptive point estimates; shared resolution and prompts do not equate the models' pretraining or optimization procedures.

Dataset	Model	Dice	IoU	HD95	CBL
BTXRD	SAM-Med2D zero-shot	0.255	0.156	41.2	0.734
	SAM-Med2D fine-tuned	0.630	0.497	17.3	0.836
	PGA-UNet 256	**0.762**	**0.633**	**10.9**	**0.902**
FracAtlas	SAM-Med2D zero-shot	0.262	0.156	**21.3**	0.752
	SAM-Med2D fine-tuned	0.563	0.417	**8.0**	0.846
	PGA-UNet 256	**0.629**	**0.470**	10.6	**0.895**

because their scoring frames differ from the shared 256×256 frame used above; each is scored in its own frame, network input and scoring frame are reported separately, and these numbers are not used to establish a cross-model ranking. The change from covering to off-center prompts is discussed separately in Section V-F.

E. DESCRIPTIVE ANALYSIS OF IMAGES WITH LOWER TOTAL LESION AREA

The lower-lesion-area image subset contains the 50 test images per dataset with the smallest total merged lesion-mask area in the common reference frame, ranging from 7 to 433 pixels on BTXRD and 145 to 944 pixels on FracAtlas. This image-level definition covers 26.7% of BTXRD's test images and 69.4% of FracAtlas's (Table 2) and does not measure per-lesion sensitivity or represent matched area quantiles across the two datasets. Combined with the Attention U-Net top- and bottom-Dice subsets of Section V-C, which are defined by that baseline's own predictions rather than an external difficulty criterion, these analyses describe the selected images and are not independent evidence that the proposed method preferentially benefits small lesions or intrinsically difficult cases in general.

Tables 8 and 9 use a shared 512 × 512 scoring frame, whereas Table 10 reports results scored in each configuration's own frame. The 256-input results in Table 8 therefore differ from the corresponding 256-frame results in Table 10,

TABLE 7. Configuration-specific reference results under covering prompts, image-level merged. Network input resolution and metric scoring frame are reported separately. Scores from different frames are retained to document the evaluated configurations and are not used to establish a cross-model ranking; raw HD95 values from different scoring frames are not directly comparable and are left unbolded throughout.

Dataset	Model	Input / frame	Dice	IoU	HD95	CBL
BTXRD	PGA-UNet	512 / 512	0.782	0.661	22.8	0.918
	PGA-UNet	128 / 128	0.714	0.572	5.8	0.859
	MedSAM zero-shot	1024 / orig.	0.304	0.185	242.4	0.832
	MedSAM fine-tuned	1024 / orig.	0.724	0.590	79.9	0.883
	ScribblePrompt zero-shot	128 / orig.	0.411	0.285	232.2	0.774
	ScribblePrompt fine-tuned	128 / orig.	0.653	0.514	122.7	0.853
FracAtlas	PGA-UNet	512 / 512	0.727	0.585	10.5	0.913
	PGA-UNet	128 / 128	0.596	0.445	8.1	0.849
	MedSAM zero-shot	1024 / orig.	0.319	0.193	60.1	0.843
	MedSAM fine-tuned	1024 / orig.	0.729	0.583	16.2	0.900
	ScribblePrompt zero-shot	128 / orig.	0.432	0.290	34.2	0.832
	ScribblePrompt fine-tuned	128 / orig.	0.679	0.527	17.9	0.891

TABLE 8. Small-lesion subset, PGA-UNet vs SAM-Med2D at the matched 256 $\times$ 256 resolution. 50 images per dataset, covering prompts, scored in a shared 512 $\times$ 512 frame.

Dataset	Model	Dice	IoU	HD95	CBL
BTXRD	SAM-Med2D zero-shot 256	0.139	0.082	62.4	0.515
	SAM-Med2D fine-tuned 256	0.256	0.170	123.8	0.561
	PGA-UNet 256	**0.705**	**0.559**	**14.7**	**0.877**
FracAtlas	SAM-Med2D zero-shot 256	0.205	0.123	33.5	0.659
	SAM-Med2D fine-tuned 256	0.371	0.256	62.1	0.736
	PGA-UNet 256	**0.648**	**0.491**	**20.6**	**0.897**

even though both use the same 50-image subset. This is a separate distinction from the fact that Table 3 additionally evaluates a different image set, the full test split rather than this 50-image subset, so its numbers are not compared here as though frame were the only difference.

a: Against SAM-Med2D at the matched resolution

SAM-Med2D struggles clearly at this lesion size (Table 8). Even after fine-tuning at 256, it reaches only 0.256 Dice on BTXRD and 0.371 on FracAtlas; in zero-shot mode it reaches 0.139 and 0.205. PGA-UNet at the same 256 resolution still holds 0.705 and 0.648, with far lower HD95.

b: Against prompt-matched conventional baselines at 512

Given the same box at 512, the two conventional prompt baselines trail PGA-UNet by about 0.04 on BTXRD and by 0.11 to 0.16 on FracAtlas, while prompt-free Attention U-Net drops to only about 0.28 (Table 9). The prompt-crop variant reaches a lower HD95 on BTXRD, perhaps because its tight crop suits a compact target well, but this does not carry over to FracAtlas.

c: Resolution on the small-lesion images

This same fixed set of 50 images per dataset is held constant and evaluated in turn at 128, 256, and 512; predictions and ground truth are merged at image level at each resolution, HD95 is reported both in native pixels and in normalized form, and the degradation is measured relative to the 512 result. On these images, PGA-UNet's Dice falls from 0.766 at 512 to 0.631 at 128 on BTXRD, and from 0.720 to 0.609

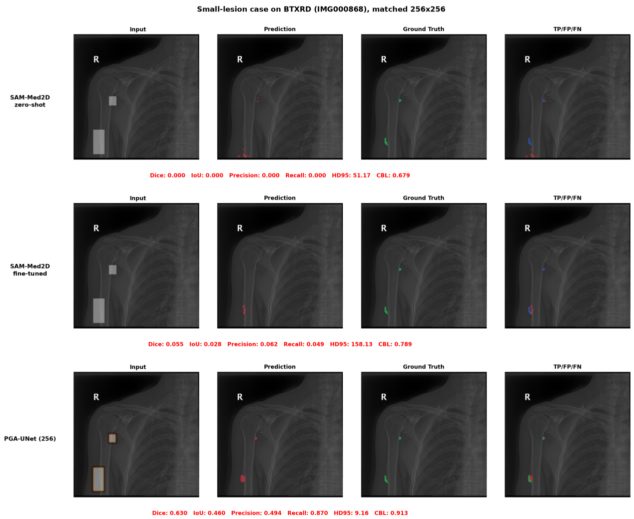

FIGURE 5. Small-lesion case on BTXRD (IMG000868, two GT polygons): SAM-Med2D zero-shot, SAM-Med2D fine-tuned, and PGA-UNet, all at 256 $\times$ 256. Both SAM-Med2D variants miss the lesions almost entirely (Dice 0.000, 0.055); PGA-UNet reaches 0.630.

TABLE 9. Small-lesion subset at 512 $\times$ 512, PGA-UNet vs the two conventional prompt baselines. 50 images per dataset, covering prompts.

Dataset	Model	Dice	IoU	HD95	CBL
BTXRD	Attention U-Net (no prompt)	0.283	0.219	178.2	0.283
	U-Net + binary prompt channel	0.728	0.591	14.2	0.879
	Attention U-Net + prompt crop	0.724	0.602	**5.5**	**0.909**
	PGA-UNet	**0.766**	**0.637**	14.0	0.897
FracAtlas	Attention U-Net (no prompt)	0.291	0.216	139.4	0.401
	U-Net + binary prompt channel	0.610	0.449	21.9	0.888
	Attention U-Net + prompt crop	0.564	0.425	22.6	0.855
	PGA-UNet	**0.720**	**0.578**	**8.3**	**0.907**

on FracAtlas; the loss from 512 to 128 is 0.135 and 0.111 respectively (Table 10). On BTXRD this loss is larger than the 0.068 lost over the same resolution range on the full test set (Table 3); on FracAtlas the full-test loss is itself large (0.131), so the loss on this subgroup is comparable rather than larger there. As with the full-resolution comparison above, each configuration is scored in its own frame, so this combines resolution and mask-sampling effects rather than isolating either.

On this lower-lesion-area image subset, PGA-UNet's margin over the no-prompt baseline remains large on both datasets, and the resolution effect is larger than on the full test set on BTXRD; on FracAtlas the full test set already loses this much to resolution, so the subgroup does not show a further resolution penalty there. As noted above, this subset is a descriptive selection of images and does not by itself establish that the method preferentially benefits small lesions in general.

F. SENSITIVITY TO PROMPT DISPLACEMENT

Under the predefined off-center construction, PGA-UNet's Dice decreases by 0.008 on BTXRD and 0.014 on FracAtlas (Table 11); on the 50-image lower-lesion-area image subset of

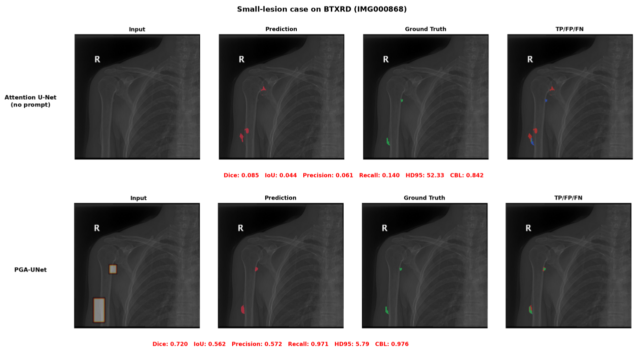

FIGURE 6. **Small-lesion case on BTXRD (IMG000868, two GT polygons), Attention U-Net (no prompt) vs PGA-UNet, both at 512 $\times$ 512. Without a prompt, the automatic baseline all but misses both lesions (Dice 0.085); with a covering box prompt, PGA-UNet recovers them (Dice 0.720). The two conventional prompt baselines are quantified in Table 9 but omitted here for readability.**

TABLE 10. **PGA-UNet on the fixed 50-image lower-lesion-area image subset of each dataset across resolutions. Dice, IoU, and CBL are reported as covering / off-center; each configuration is scored in its own frame. The HD95 pair denotes native pixels / normalized distance under the covering condition. The Delta Dice column is the covering-prompt difference relative to the 512 configuration.**

Dataset	Res.	Dice	IoU	CBL	HD95$_{\text{px}}$ / $_n$	ΔDice
BTXRD	128	0.631 / 0.662	0.488 / 0.517	0.738 / 0.758	3.9 / 0.022	-0.135
	256	0.714 / 0.695	0.570 / 0.548	0.869 / 0.849	7.2 / 0.020	-0.052
	512	**0.766 / 0.757**	**0.637 / 0.627**	**0.897 / 0.890**	14.0 / 0.019	—
FracAtlas	128	0.609 / 0.543	0.462 / 0.393	0.828 / 0.757	8.9 / 0.049	-0.111
	256	0.642 / 0.610	0.484 / 0.451	0.887 / 0.845	10.3 / 0.029	-0.079
	512	**0.720 / 0.712**	**0.578 / 0.570**	**0.907 / 0.893**	8.3 / 0.012	—

Section V-E the decrease stays under 0.01 for both datasets. The binary channel baseline has a smaller decrease on FracAtlas (0.005), so the results do not establish that PGA-UNet is uniformly the least sensitive model to this construction; the crop baseline decreases more on FracAtlas (0.027) and about as much on BTXRD (0.008). PGA-UNet retains the highest observed Dice among the three configurations in this table under both prompt conditions on both datasets. These observations apply to the evaluated lesion-covering box construction and do not characterize arbitrary user-drawn, partial-coverage, or off-target prompts.

For reference, the no-prompt Attention U-Net is by construction unaffected by this box displacement (0.517 on BTXRD, 0.342 on FracAtlas under both conditions), and the three fine-tuned promptable reference models, each scored in its own native frame, lose 0.03 to 0.08 Dice under the same off-center construction (SAM-Med2D: -0.029 BTXRD / -0.031 FracAtlas; MedSAM: -0.033 / -0.063; ScribblePrompt-UNet: -0.080 / -0.081); these three are not included in Table 11 because their scoring frames differ from the shared 512 $\times$ 512 frame used there.

G. ABLATION STUDY

The ablation drops one factor at a time from the full model (Gaussian prompt combined with PSG and CAD), and separately tests replacing the Gaussian prompt with a binary box

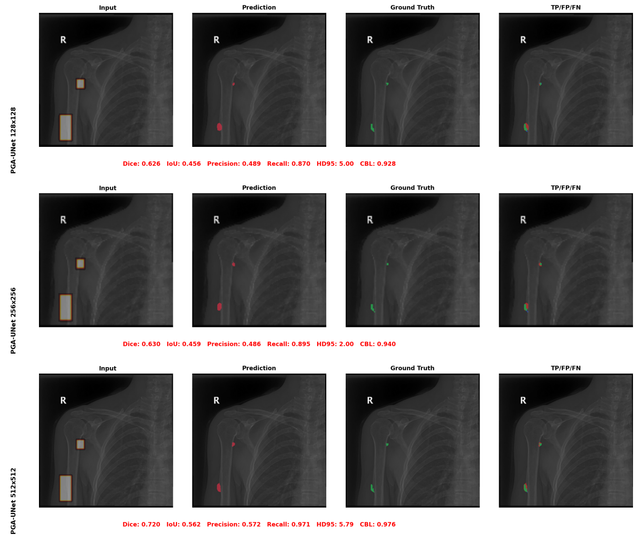

FIGURE 7. **PGA-UNet on the same small-lesion case on BTXRD (IMG000868, two GT polygons) at 128 $\times$ 128, 256 $\times$ 256, and 512 $\times$ 512 (Dice 0.626, 0.630, 0.720). The prediction tightens around the lesion as resolution increases.**

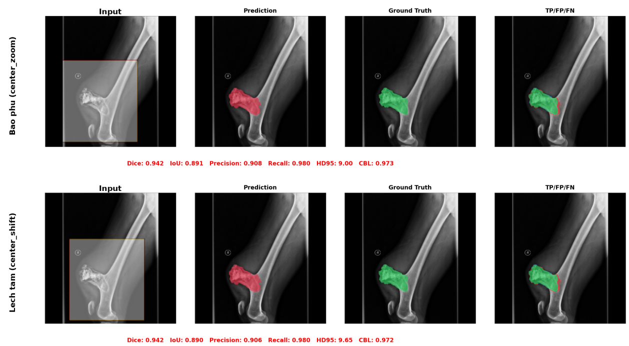

FIGURE 8. **PGA-UNet on the same BTXRD case (IMG000020) under a covering prompt (top) and an off-center prompt (bottom); Dice is unchanged at 0.942.**

(Table 12). The full model is first or second in all four dataset-and-prompt combinations; each ablated variant is top-two in exactly one. The binary-box variant leads BTXRD covering by 0.006 Dice but trails by 0.059 on FracAtlas, and PSG-only is close to the full model on BTXRD yet falls 0.092 Dice below it on FracAtlas. These are single-split results.

The component effects depend on the dataset. On FracAtlas, the full configuration has higher Dice than the CAD-removed and binary-prompt variants, and these differences survive the reported Holm correction over the full family of ablation tests (Table 13); the CAD-removed variant loses 0.09 to 0.10 Dice. On BTXRD, where every configuration already sits near 0.78, the reported Dice differences from the ablated variants do not survive that correction, including the 0.006 Dice binary-box lead (bootstrap 95% CI $[-0.017, +0.004]$); the binary-prompt variant also has a higher covering-prompt point estimate than the Gaussian configuration there. The ablations therefore support dataset-dependent benefits of the

TABLE 11. Observed sensitivity to the predefined off-center box construction (z: covering, s: off-center). All configurations use a 512 $\times$ 512 input and scoring frame; crop predictions are mapped back to the full frame. The Delta Dice column is off-center minus covering Dice. Boxes preserve full lesion coverage, and the off-center construction may change box extent when needed to retain coverage. Results are descriptive point estimates on the fixed test splits.

Dataset	Model	Dice z/s	ΔDice	IoU z/s	HD95 z/s	CBL z/s
BTXRD	U-Net + binary prompt channel	0.760 / 0.748	−0.012	0.635 / 0.626	33.0 / 32.6	0.895 / 0.890
	Attention U-Net + prompt crop	0.741 / 0.733	−0.008	0.615 / 0.607	24.7 / 26.4	0.912 / 0.896
	PGA-UNet	**0.782 / 0.774**	−0.008	**0.661 / 0.652**	**22.8** / 24.5	**0.918 / 0.912**
FracAtlas	U-Net + binary prompt channel	0.593 / 0.588	**−0.005**	0.434 / 0.428	22.9 / 24.0	0.887 / 0.860
	Attention U-Net + prompt crop	0.590 / 0.563	−0.027	0.451 / 0.428	22.0 / 23.8	0.865 / 0.833
	PGA-UNet	**0.727 / 0.713**	−0.014	**0.585 / 0.570**	**10.5 / 10.4**	**0.913 / 0.897**

TABLE 12. Ablation at 512 $\times$ 512, image-level merged, single split. Dice, IoU, HD95 and CBL as covering / off-center.

Dataset	Configuration	Dice	IoU	HD95	CBL
BTXRD	CAD only	0.771 / 0.759	0.649 / 0.635	23.1 / 24.2	0.915 / 0.905
	PSG only	0.779 / 0.771	0.660 / 0.651	29.4 / 28.4	0.908 / 0.902
	PSG + vanilla attention gate	0.769 / 0.757	0.645 / 0.633	26.4 / 28.7	0.906 / 0.896
	Full model + binary box prompt	**0.788** / 0.770	**0.668** / 0.651	23.1 / 28.5	0.914 / 0.900
	Full PGA-UNet (Gaussian + PSG + CAD)	0.782 / **0.774**	0.661 / **0.652**	**22.8 / 24.5**	**0.918 / 0.912**
FracAtlas	CAD only	0.696 / 0.685	0.551 / 0.540	18.0 / **12.0**	0.903 / 0.885
	PSG only	0.635 / 0.611	0.479 / 0.459	14.5 / 14.8	0.894 / 0.857
	PSG + vanilla attention gate	0.690 / 0.691	0.548 / 0.547	14.0 / 14.1	0.892 / 0.881
	Full model + binary box prompt	0.668 / 0.657	0.514 / 0.505	12.0 / 13.2	0.904 / 0.886
	Full PGA-UNet (Gaussian + PSG + CAD)	**0.727 / 0.713**	**0.585 / 0.570**	**10.5** / 10.4	**0.913 / 0.897**

TABLE 13. Paired comparison of the full model against each ablated variant, per-image Dice on the single split. The Delta column is the mean per-image Dice difference (full minus variant), tested with a two-sided paired sign-flip randomization test (20,000 permutations). ** marks entries that stay significant ($p < 0.05$) after Holm correction over the full family of ablation tests (four ablated model contrasts $\times$ two datasets $\times$ two prompt conditions, all on Dice, the only metric tested here); * marks raw $p < 0.05$ that does not survive correction.

ΔDice, full vs variant	BTXRD		FracAtlas	
	covering	off-center	covering	off-center
CAD only (no PSG)	$+0.010$	$+0.015^{*}$	$+0.031^{*}$	$+0.028^{*}$
PSG only (no CAD)	$+0.002$	$+0.003$	$+0.092^{**}$	$+0.102^{**}$
PSG + vanilla attention gate	$+0.012$	$+0.017^{*}$	$+0.038^{*}$	$+0.022$
Full + binary box prompt	-0.006	$+0.004$	$+0.059^{**}$	$+0.056^{**}$

complete conditioning design, without establishing that every component is necessary in every setting. They do not directly measure preservation of weak lesion features or estimate variation across training seeds, and a single split could not settle finer, component-by-component contributions in any case.

H. MONTE CARLO CROSS-VALIDATION

We retrained PGA-UNet at 512 $\times$ 512 on four additional random image-level splits, keeping the training seed fixed at 22120196 and changing only the split seed (Table 14). Across these splits, PGA-UNet obtains covering-prompt Dice of 0.769 ± 0.005 on BTXRD and 0.733 ± 0.007 on FracAtlas, and off-center Dice of 0.762 ± 0.004 and 0.728 ± 0.011, respectively (mean $\pm$ standard deviation); CBL varies with a standard deviation of at most 0.013. These results characterize PGA-UNet's own observed split-to-split variation; they do not establish comparable variation for the baselines, which were not retrained across these splits, or the stability of the differences from competing models reported elsewhere in this article.

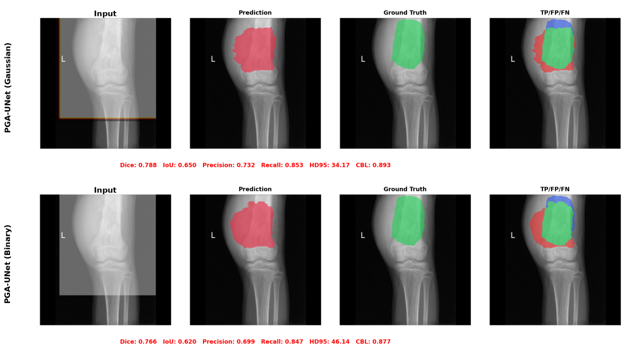

FIGURE 9. Full PGA-UNet on the same BTXRD case (IMG000054) with a Gaussian-plateau prompt vs a binary box prompt, both at 512 $\times$ 512. The binary variant loses localization even though PSG and CAD are unchanged.

TABLE 14. Monte Carlo cross-validation of PGA-UNet at 512 $\times$ 512, four repeated random image-level splits, mean $\pm$ standard deviation.

Dataset	Prompt	Dice	IoU	HD95	CBL
BTXRD	Covering	0.769 ± 0.005	0.647 ± 0.006	25.0 ± 2.5	0.905 ± 0.005
	Off-center	0.762 ± 0.004	0.639 ± 0.007	25.7 ± 2.0	0.901 ± 0.007
FracAtlas	Covering	0.733 ± 0.007	0.595 ± 0.007	14.1 ± 2.7	0.909 ± 0.006
	Off-center	0.728 ± 0.011	0.590 ± 0.010	15.6 ± 3.9	0.904 ± 0.013

I. TRAINING-SEED STABILITY ANALYSIS

The Monte Carlo cross-validation above varies the image-level split while keeping the training seed fixed at 22120196.

To characterize the complementary source of variability, training-run randomness itself, we are separately retraining PGA-UNet at 512×512 on the original fixed split (Table 2) using three additional training seeds, with every other setting from Section IV-B held fixed. All four runs use the same fixed train/validation/test partition; only the seed governing weight initialization, data shuffling, and other stochastic training behavior changes between them.

This analysis is restricted to PGA-UNet. None of the evaluated baselines, the two conventional prompt baselines or the three promptable foundation models, has been retrained across multiple seeds, so no comparison in this article claims stable superiority over those baselines across training seeds. Of the three additional seeds, one (19122003) has completed on both datasets at the time of this draft; the other two are still training. Table 15 therefore reports an interim mean $\pm$ sample standard deviation over the $n = 2$ completed seeds (22120196 and 19122003), not the full four; the sample size is stated in the caption, and the table will be extended with the remaining two seeds, not recomputed from scratch, once they complete (see `PENDING_MULTISEED.md`).

With only two completed seeds, these standard deviations are not yet informative about the true seed-to-seed spread and should not be read as a stable estimate; they describe only the two runs observed so far. Read as a two-point comparison, not a distribution: the newly completed seed's Dice is about 0.02 lower than the original seed's on BTXRD under both prompt conditions, while on FracAtlas the two seeds are within 0.002 Dice of each other. Between the two completed seeds, Precision differs by about 0.034 on BTXRD under both prompt conditions and by 0.010 to 0.012 on FracAtlas, while Recall differs by 0.001 to 0.002 on BTXRD and by 0.006 to 0.016 on FracAtlas. HD95 differs by about 1 to 2 pixels on BTXRD covering and by about 1 pixel on FracAtlas off-center, and by under 1 pixel on the other two rows. The picture is not uniform across metrics: Dice and Precision differ more between the two seeds on BTXRD than on FracAtlas, while Recall differs less on BTXRD than on FracAtlas. Whether any of these patterns hold up is exactly what the two remaining seeds will show; nothing here yet supports a general claim about which dataset or metric is more sensitive to the training seed.

J. EFFICIENCY

PGA-UNet contains approximately 2.96 million parameters (Table 16), about 92 times fewer than fine-tuned SAM-Med2D's 271.24 million. In the reported Tesla T4, batch-size-one measurement, a single forward pass takes 8.7 ms at 256×256 and 18.5 ms at 512×512; the evaluated SAM-Med2D configuration takes 50.3 ms at 256×256. ScribblePrompt-UNet takes 6.2 ms at its 128×128 input, with 3.99 million parameters, a closer parameter count to PGA-UNet's than SAM-Med2D's; MedSAM's 1024×1024 input gives it the highest GFLOPs and latency in the group even though it has far fewer parameters than SAM-Med2D. Quality at these same per-model resolutions is reported separately in Tables 6 and 7; a smaller parameter count or lower latency here does not by itself imply better segmentation, and those tables should be read together, not this one alone, when weighing a model choice. Latency is measured in FP32 with synthetic random-tensor inputs at each model's own resolution, not real images; each GPU figure is the mean over 50 timed forward passes after 10 warmup passes, with `torch.cuda.synchronize()` called after warmup and after every timed pass, and each CPU figure is the mean over 20 timed passes after 5 warmup passes on the same machine. The specific CPU model was not captured by the benchmark script and is not available to report. These measurements describe the tested implementations on one specific GPU and do not rank complete interactive workflows. They do not include a separate analysis of first-prompt versus cached subsequent-prompt operation, which would matter for the SAM-family models since they can reuse one image embedding across several prompts on the same image, and a single forward-pass time is not the latency of the perturbation-based self-assessment score or candidate shortlist, which each require several forward passes.

K. EXPLORATORY COMPARISON OF SEGMENTATION LOSSES

Because the segmentation targets in this article are small, we tried replacing the Dice term with a size-conditioned Tversky loss, and in another trial, with a Focal Dice loss, with every other setting held fixed (Table 17). Neither option worked better on Dice, which is best or tied for the default Dice + BCE objective on both datasets and both prompt conditions; HD95 follows the same pattern. CBL mostly favors the default too, with one exception: on FracAtlas covering, Tversky's CBL (0.918) is slightly higher than the default's (0.913), even though its Dice, IoU, and HD95 are all worse there. Focal Dice is a clear step backward on FracAtlas on every metric, with HD95 exceeding 100 pixels. We treat this as a negative control: under the current protocol, the default objective remains the better choice on the metric it is trained to optimize, and neither alternative is adopted.

L. FAILURE MODES

PGA-UNet still makes mistakes, and these mistakes are not random but recur in a few patterns. In qualitative review, low Dice tends to cluster on lesions that are elongated or branching, lesions located in cluttered anatomy such as the shoulder or clavicle, or lesions that barely stand out from the surrounding bone. In such cases, even a good covering box does not help if the local image evidence itself is too weak.

M. TRAINING-FREE SELF-ASSESSMENT AND CANDIDATE SUGGESTION

The two support signals below are computed directly from the trained model's output and are reported as preliminary. Q (Section III-F) is computed without ground-truth masks at inference; on the annotated lesion-bearing test images, its association with per-prompt Dice is quantified by the

TABLE 15. Training-seed stability of PGA-UNet at 512 × 512 on the original fixed split (Table 2). Interim results from $n = 2$ completed training seeds (22120196 and 19122003) of the four planned; mean ± sample standard deviation (ddof = 1). Two additional seeds are still training and are not included here. Precision and recall are the standard pixel-level values for the image-level merged mask, reported only in this table; they are not a lesion-level detection recall, which this article does not measure. See `PENDING_MULTISEED.md`.

Dataset	Prompt	Dice	IoU	Precision	Recall	HD95	CBL
BTXRD	Covering (`center_zoom`)	0.771 ± 0.016	0.645 ± 0.022	0.745 ± 0.024	0.831 ± 0.000	23.6 ± 1.2	0.914 ± 0.005
	Off-center (`center_shift`)	0.762 ± 0.017	0.637 ± 0.021	0.747 ± 0.024	0.819 ± 0.002	24.8 ± 0.5	0.907 ± 0.007
FracAtlas	Covering (`center_zoom`)	0.728 ± 0.001	0.586 ± 0.002	0.678 ± 0.008	0.835 ± 0.011	10.6 ± 0.1	0.908 ± 0.006
	Off-center (`center_shift`)	0.713 ± 0.001	0.570 ± 0.000	0.674 ± 0.007	0.806 ± 0.004	11.0 ± 0.8	0.899 ± 0.002

TABLE 16. Parameters, FLOPs, and single forward-pass latency in the tested environment (NVIDIA Tesla T4, batch size 1), one BTXRD checkpoint per model.

Model	Params	GFLOPs	GPU ms	CPU ms
PGA-UNet 256	2.96M	7.74	8.7	93.6
PGA-UNet 512	2.96M	30.97	18.5	341.7
SAM-Med2D 256	271.24M	92.02	50.3	1343.6
MedSAM 1024	93.74M	976.48	381.7	12129.4
ScribblePrompt 128	3.99M	32.81	6.2	187.4

TABLE 17. Exploratory segmentation-loss comparison for PGA-UNet at 512 × 512, everything else fixed. Dice, IoU, HD95, and CBL as covering / off-center. The default is not changed.

Dataset	Loss	Dice	IoU	HD95	CBL
BTXRD	Dice + BCE (default)	**0.782 / 0.774**	**0.661 / 0.652**	22.8 / 24.5	**0.918 / 0.912**
	Size-conditioned Tversky	0.778 / 0.767	0.660 / 0.649	25.2 / 30.3	0.916 / 0.903
	Focal Dice	0.773 / 0.765	0.649 / 0.642	25.6 / 25.6	0.913 / 0.907
FracAtlas	Dice + BCE (default)	**0.727 / 0.713**	**0.585 / 0.570**	**10.5 / 10.4**	0.913 / **0.897**
	Size-conditioned Tversky	0.709 / 0.680	0.561 / 0.530	11.0 / 12.2	**0.918** / 0.888
	Focal Dice	0.642 / 0.578	0.493 / 0.425	105.5 / 110.1	0.802 / 0.732

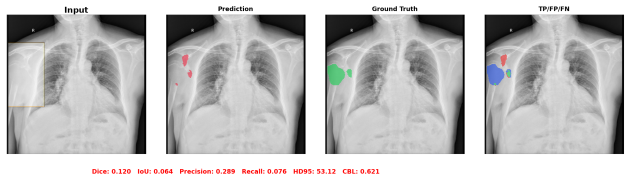

FIGURE 10. Failure case on BTXRD (IMG000013, Dice 0.120): the prompt covers the lesion at the chest/rib margin, but PGA-UNet's prediction lands away from it. A covering box does not fix a case where the local image evidence is itself ambiguous.

TABLE 18. Spearman correlation between each training-free cue and the true per-prompt Dice. $Q = \text{mean}(S_{\text{sharp}}, S_{\text{stab}})$.

	BTXRD		FracAtlas	
Cue	cover	off-center	cover	off-center
S_{sharp} (mask decisiveness)	0.42	0.39	0.47	0.43
S_{stab} (jitter stability)	0.64	0.56	0.63	0.45
Q	**0.70**	**0.60**	**0.65**	**0.48**

correlations below (Table 18). Each lesion polygon (one prompt) is treated as an independent unit in this correlation, permutation, and bootstrap analysis, at the sample sizes in Table 2 (e.g. 232 BTXRD test polygons); images with more than one lesion contribute one unit per polygon rather than one unit per image, and we have not adjusted the analysis for this within-image dependence. The Spearman correlation between Q and the true Dice reaches 0.70 (covering) and 0.60 (off-center) on BTXRD, and 0.65 and 0.48 on FracAtlas; all four are significantly different from zero (permutation test, $p < 0.001$), with bootstrap 95% intervals ranging from $[0.61, 0.77]$ on BTXRD covering to $[0.29, 0.63]$ on FracAtlas off-center. Pearson correlation follows the same pattern (0.82 and 0.76 on BTXRD, 0.59 and 0.44 on FracAtlas); we report Spearman as the primary statistic since it does not assume a linear Q-Dice relationship. Both component cues contribute meaningfully: mask decisiveness alone already reaches a Spearman of nearly 0.4, jitter stability reaches about 0.5 to 0.6, and averaging the two cues is always at least as good as either alone. As a control, we also tried training a small head to directly regress the Dice; in this specific experiment it collapsed to a near-constant output, with Spearman under 0.2 on both datasets, so we did not adopt it here. This single comparison does not establish that a learned confidence head cannot work in general, only that this particular configuration did not.

Used in the reverse direction, Q can also rank 50 randomly sampled candidate boxes and keep only the best few (Table 19, Figure 11). The best of the top five boxes covers 0.84 and 0.90 of the lesion area on average on BTXRD and FracAtlas, and reaches near-complete coverage (≥ 0.999) for 67% and 82% of images respectively. To check that this ranking earns its keep, we compare against two baselines computed on the same sampled 50-box pool per image: a random-5 selection (mean of 2,000 resamples per image) and an oracle-5 that picks the 5 truly best-covering boxes, an upper bound no ranking rule could see without the ground truth. Q-ranking roughly doubles mean coverage over random selection (0.52 vs 0.23 on BTXRD, 0.62 vs 0.26 on FracAtlas) and raises the near-complete-coverage rate by 26 to 27 points (67% vs 40% on BTXRD, 82% vs 55% on FracAtlas), closing much of the gap to the oracle's 86% and 89%. This is only a scaffold for a small suggestion feature: Q is not a true probability, the shortlist does not replace the clinician's own review, and as the next paragraph shows, it is only meaningful when a lesion is actually present.

Q presupposes that a lesion lies inside the box. To probe this assumption, we ran the same 50-box sampling on the lesion-free images of each dataset (1879 tumor-free BTXRD images, 3366 fracture-free FracAtlas images), where no coverage is defined (Table 20). Q does not stay reliably low: the

TABLE 19. Candidate-region shortlist: Q-ranking vs a random-5 baseline and an oracle-5 upper bound, all drawn from the same 50 prompt-free boxes sampled per image. Coverage is the fraction of lesion pixels inside a box; near-complete-cov. counts images whose best-of-5 box reaches ≥ 0.999 coverage (random-5 uses the mean over 2,000 resamples per image).

Dataset	Method	Mean cov.	Best-of-5 cov.	Near-complete-cov.
BTXRD	Q-ranked	**0.52**	0.84	66.8%
	Random-5	0.23	0.67	40.4%
	Oracle-5	—	**0.98**	**86.1%**
FracAtlas	Q-ranked	**0.62**	0.90	81.9%
	Random-5	0.26	0.74	55.0%
	Oracle-5	—	**0.97**	**88.9%**

TABLE 20. Distribution of the self-assessment score Q on lesion-free images, with 50 random boxes sampled per image. With no lesion present, coverage is undefined; only the distribution of Q is reported.

Dataset	Lesion-free img.	Median Q	p90 Q	Boxes $Q \geq 0.5$	$\overline{Q_{\max}/\text{img}}$
BTXRD	1879	0.00	0.75	28%	0.85
FracAtlas	3366	0.07	0.82	42%	0.90

median collapses toward zero, but about 28% of boxes on BTXRD and 42% on FracAtlas still exceed $Q = 0.5$. The per-image maximum Q averages 0.85 on BTXRD and 0.90 on FracAtlas. Together with the reported proportions of high-scoring boxes, these results show that high Q values can occur on lesion-free anatomy. Q should therefore not be interpreted as evidence of lesion presence: it ranks how good a mask is once a lesion has been located, and does not by itself flag lesion absence. This pattern is consistent with the model's weights being learned only on lesion-bearing images, with no negative class. Adding lesion-free images to training is a direction we plan to develop.

a: Exploratory post-hoc gate: MedSigLIP

As one exploratory response to the finding above, we tested whether a separate, off-the-shelf vision-language model could catch some of the false-positive boxes that Q misses on lesion-free images. This check uses BTXRD only, not FracAtlas, and reuses the same trained PGA-UNet 512×512 checkpoint as the rest of this article. We scored 50 sampled boxes per image with MedSigLIP [18], a medical image-text encoder, used zero-shot as a post-hoc gate applied after PGA-UNet's prediction, not as part of the architecture and not fine-tuned. A fixed ensemble of positive and negative text prompts yields a cosine-margin score, thresholded at a cutoff chosen only on a validation split of 410 lesion-free images to retain 95% of positive validation boxes; the held-out test images play no part in choosing this threshold. The held-out test split is the remaining 1,469 of the same 1,879 lesion-free BTXRD images used above, giving 73,450 sampled boxes.

Gating with MedSigLIP lowers the box-level false-positive rate on these held-out boxes from 61.96% to 44.62%, an absolute reduction of 17.33 points (paired cluster-bootstrap 95% CI $[16.68, 17.98]$; Table 21). On the article's own BTXRD test-split lesion boxes (232 boxes from 187 images, built

TABLE 21. MedSigLIP zero-shot post-hoc gate on BTXRD lesion-free images (held-out test split, 1,469 images, 50 boxes/image, 73,450 boxes). Positive-box retention is measured separately on the article's own BTXRD test-split lesion boxes (232 boxes, 187 images). Exploratory, not a core contribution; no post-gate Dice is reported.

Quantity	Before gate	After gate
Box FPR (73,450 boxes)	61.96%	44.62%
Image FPR (1,469 images)	100%	99.93%
Positive-box retention (232 boxes)	96.12%	

with the same covering-box construction used throughout this article), the gate retains 96.12% of positive boxes (cluster-bootstrap 95% CI $[93.42, 98.30]$). Image-level false-positive rate, the fraction of lesion-free images with at least one surviving false-positive box among the 50 sampled, barely moves, from 100% to 99.93%: nearly every lesion-free image still has at least one box that fools the gated system. We do not report a post-gate Dice or other segmentation-quality measure here. This result is exploratory and supplementary, not a core contribution of this article: a lightweight external gate can reduce box-level false responses while keeping most positive boxes, without solving image-level false-positive detection; developing a purpose-built, better-calibrated lightweight gate is future work (Section VI).

VI. DISCUSSION

The principal finding is that a compact network with explicit box conditioning achieves higher observed Dice than the tested conventional prompt baselines under the controlled protocol. This is supported by the comparisons at a shared scoring frame, including the binary-prompt PGA-UNet variant. The Gaussian full model and the binary channel baseline nevertheless differ in both representation and architecture, so their difference cannot be assigned entirely to PSG or CAD. The component ablations further indicate that the measured benefits depend on the dataset. In particular, the binary prompt remains competitive on BTXRD.

These experiments evaluate segmentation after a lesion has already been localized. The simulated boxes retain full coverage and encode lesion extent through their construction from annotations. The results therefore concern this prompt distribution and do not demonstrate performance with unrestricted clinician-drawn boxes. Explicit prompt injection is a property of the architecture; preservation of weak lesion features is a design motivation that has not been directly measured.

The main baseline differences are descriptive estimates from fixed test splits, each a single fixed-run comparison rather than a repeated training run. Existing paired ablation statistics are conditional on the evaluated checkpoints, while the additional splits characterize PGA-UNet alone. A training-seed stability analysis restricted to PGA-UNet (Section V-I) currently reports two of the four planned training seeds; it does not, by itself, establish stable superiority over any baseline, none of which has been retrained across mul-

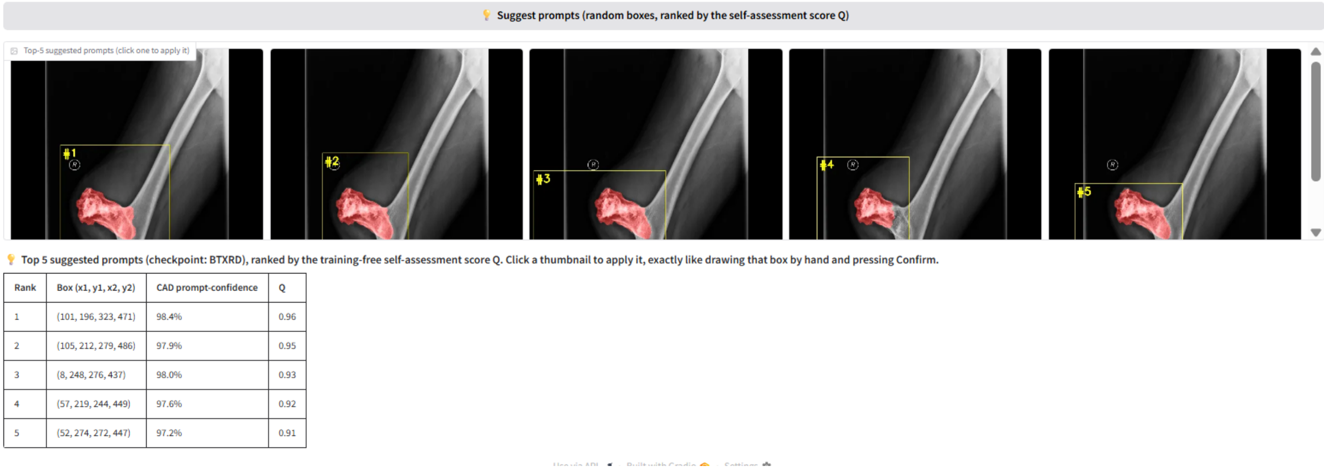

FIGURE 11. Candidate-region shortlist on a BTXRD case. Fifty boxes are sampled with no prompt, scored by the self-assessment score *Q*, and the five highest-scoring boxes are shown for the clinician to accept, adjust, or reject. A FracAtlas case is shown in the supplementary material.

tiple seeds, and its own two remaining seeds may change the observed spread. Patient-level separation is unverified, and the difference between the converted FracAtlas polygon count and the original instance count remains unreconciled at annotation-ID level. Separate models are trained for the two datasets, without cross-dataset evaluation. Accordingly, the study supports a bounded empirical assessment of the proposed design, rather than general claims about clinical utility, comparative superiority across training runs, or cross-dataset generalization.

VII. CONCLUSION

This work introduced PGA-UNet, a compact CNN that incorporates a box prompt into encoder features and decoder skip attention. On the fixed BTXRD and FracAtlas test splits, the model achieved covering-prompt Dice scores of 0.782 and 0.727 at 512×512, with higher observed Dice than the tested conventional prompt baselines. The existing binary-prompt comparison and component ablations provide evidence for the evaluated conditioning design, with effects that depend on the dataset. PGA-UNet contains approximately 2.96 million parameters, and the reported timing results characterize its computational cost in the tested environment.

These findings are limited to simulated lesion-covering prompts, image-level evaluation, and the reported training configurations. Evaluation with clinician-drawn prompts remains future work. A training-seed stability analysis for PGA-UNet (Section V-I) reports two of four planned training seeds on the original fixed split, with the remaining two still training, and repeated comparative training runs for the evaluated baselines remain future work. The exploratory MedSigLIP gate (Section V-M) shows that a lightweight post-hoc filter can reduce box-level false positives on lesion-free images while keeping most positive boxes, without resolving image-level false-positive detection; developing a purpose-built, better-calibrated lightweight gate, together with a post-gate segmentation-quality evaluation, is left for future work.

DATA AVAILABILITY

BTXRD and FracAtlas are public datasets cited in this article. The manuscript reports experiments on image-level train/validation/test splits derived from those datasets. Code and split details are released at https://github.com/ThongLuc2k3/PGA_Unet2D [14].

AUTHOR CONTRIBUTIONS

Thong Luc: conceptualization, methodology, software, investigation, formal analysis, visualization, writing of the original draft. Nguyen Huu Binh: software support, data preparation, experiment verification, review and editing. Ly Quoc Ngoc: supervision, validation, review and editing.

ETHICS STATEMENT

This study used public de-identified radiograph datasets and did not involve prospective data collection, patient contact, or intervention.

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